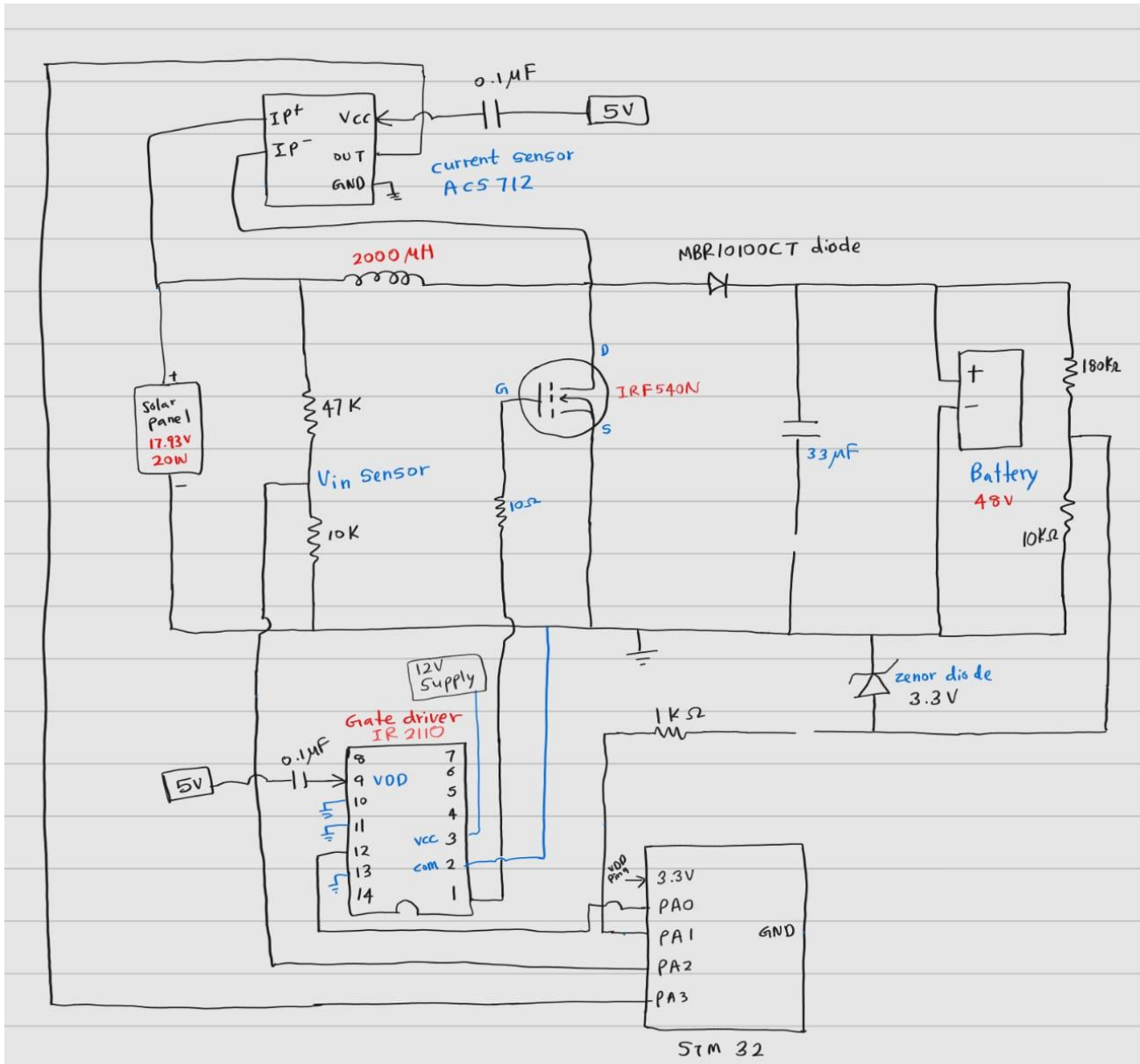


Circuit Diagram for the Prototype



Solar Panel (17.93V, 20W)

Panel (+) → ACS712 IP+ → ACS712 IP- → Top rail → L1 & Vin sensor →
 Panel (-) → Common ground

ACS712 Current Sensor

5V rail → VCC → ACS712 → OUT → STM32 ADC (PA3)

Inductor L1 (2000µH)

Top rail (from ACS712 IP-) → L1 → Switching node (MOSFET Drain + Diode Anode)

Vin Sensor Divider (47kΩ / 10kΩ)

Top rail → 47kΩ → Divider tap → 10kΩ → Common ground

MOSFET Q1 (IRF540N)

IR2110 LO (pin 1) → 10Ω gate resistor → Gate → Drain → switching node → Source → common ground

Diode D1 (MBR10100CT)

Switching node (Anode) → D1 → Cathode → Output rail

Output Capacitor (33μF)

Output rail → C1 (+) → C1 (-) → Common ground

Battery (48V)

Output rail → Battery (+) → Battery (-) → Common ground

Battery Sense Divider (180kΩ / 10kΩ) + Zener + 1kΩ

Battery (+) → 180kΩ → Divider tap → Zener 3.3V (clamp) & 10kΩ → ground → 1kΩ → STM32 ADC pin

Gate Driver IR2110

5V rail → VDD (pin 9) → — → 12V rail → VCC (pin 3) → — → COM (pin 2) → power ground

STM32 Bluepill

PA0 ← ACS712 OUT → PA1 ← Vin sensor tap → PA2 → IR2110 LIN → PA3 ← Battery sense (via 1kΩ) → GND ↔ common ground